DAILY ACTIVE RECALL · MASTER MATRIX & FACTS

# GENERAL SCIENCE & TECHNOLOGY

## सामान्य विज्ञान एवं प्रौद्योगिकी — उच्च-प्राथमिकता पुनरावलोकन

Vitamins & Deficiencies, Pathogens & Diseases (Viral/Bacterial/Protozoan), Endocrine Hormones, Optics & Eye Defects, Commercial Chemicals, Alloys, ISRO Space Missions & DRDO Missiles.

**High-Yield Ratta Matrix**
Concentrated prelims facts, tables & comparisons

**Examiner Trap Eliminators**
Confused pairs and high-frequency exam traps

**15–20 Min Daily Read**
Formulated for active morning memory drills

**Exam-Aligned Coverage**
Curated strictly for UPPSC Prelims, RO/ARO & BEO

**UP-PCS & UK-PCS Examination Resource**
General Studies Paper 1 · Prelims Fast-Recall Desk

Daily Revision Master Edition · 2026
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# Daily Read — High-Yield Revision Facts (General Science & Technology)

### ★ UPPCS & UKPCS Prelims Daily Fact Dump · Active Recall Cheat-Sheet

**Exam Focus:** UPPSC Prelims (Paper 1), UKPSC Prelims (Paper 1), RO/ARO, BEO
**Frequency:** Daily morning read (15–20 mins)
**Question Types Covered:** Vitamins, Chemical Names & Deficiency Diseases, Pathogens vs Infectious Diseases (Viral, Bacterial, Protozoan), Human Physiology & Endocrine Glands, Plant Hormones, Units & Dimensions, Optics (Lenses, Mirrors, TIR, Rainbow), Sound & Electromagnetic Spectrum, Common Chemicals & Commercial Names, Alloys & Metal Compositions, Everyday pH Values, Indian Space Missions (ISRO) & Defence Tech (DRDO Missiles, Nuclear Power Plants).

## 1. Complete Vitamin Master Reference & Deficiency Diseases

| Vitamin | Chemical / Scientific Name | Fat / Water Soluble | Key Deficiency Disease / Symptoms | Primary Natural Food Sources |
|---|---|---|---|---|
| **Vitamin A** | **Retinol** | Fat-soluble | **Night blindness (Nyctalopia)**, **Xerophthalmia** (dry cornea), Keratomalacia | Carrot, Papaya, Mango, Liver, Fish oil, Golden Rice |
| **Vitamin B1** | **Thiamine** | Water-soluble | **Beri-beri** (Peripheral neuropathy / cardiac failure), Wernicke-Korsakoff syndrome | Unpolished rice, yeast, whole grains (*lost in polished rice*) |
| **Vitamin B2** | **Riboflavin** | Water-soluble | **Cheilosis** (cracking at corners of mouth), Glossitis (red swollen tongue) | Milk, eggs, liver (*gives yellow colour to milk/whey*) |
| **Vitamin B3** | **Niacin (Nicotinic acid)** | Water-soluble | **Pellagra** (4Ds: Dermatitis, Diarrhea, Dementia, Death) | Yeast, meat, fish, legumes |
| **Vitamin B5** | **Pantothenic acid** | Water-soluble | Burning feet syndrome, fatigue | Mushroom, eggs, broccoli |
| **Vitamin B6** | **Pyridoxine** | Water-soluble | Peripheral neuropathy, microcytic anemia, convulsions in infants | Cereals, bananas, meat |
| **Vitamin B7** | **Biotin (Vitamin H)** | Water-soluble | Hair loss (alopecia), dermatitis | Egg yolk, nuts, intestinal flora (*avidin in raw egg binds biotin*) |
| **Vitamin B9** | **Folic Acid (Folate)** | Water-soluble | **Megaloblastic anemia**, Neural tube defects in fetuses (Spina bifida) | Green leafy vegetables, citrus fruits |
| **Vitamin B12** | **Cyanocobalamin** | Water-soluble | **Pernicious anemia**, demyelination of spinal cord (*contains Cobalt metal*) | Meat, fish, eggs, dairy (*Absent in plant foods*) |
| **Vitamin C** | **Ascorbic acid** | Water-soluble | **Scurvy** (bleeding gums, petechiae, impaired wound healing); heat-sensitive (*destroyed by cooking*) | Aonla (highest source), Guava, Citrus fruits (Lemon, Orange) |
| **Vitamin D** | **Calciferol** ($D_2$: Ergocalciferol, $D_3$: Cholecalciferol) | Fat-soluble | **Rickets** in children (bowed legs), **Osteomalacia** in adults (*hormone-like vitamin*) | Sunlight synthesis in skin, Cod liver oil |
| **Vitamin E** | **Tocopherol** | Fat-soluble | Infertility / reproductive failure, muscle weakness, hemolytic anemia (*Beauty Vitamin / Anti-oxidant*) | Wheat germ oil (richest), sunflower seeds, almonds |

| Vitamin | Chemical / Scientific Name | Fat / Water Soluble | Key Deficiency Disease / Symptoms | Primary Natural Food Sources |
|---|---|---|---|---|
| **Vitamin K** | **Phylloquinone** ($K_1$), Menaquinone ($K_2$) | Fat-soluble | **Defective blood clotting**, prolonged bleeding time (needed for synthesis of prothrombin in liver) | Spinach, kale, synthesized by human gut bacteria |

> **Fat-Soluble vs Water-Soluble Mnemonic:**
> Fat-soluble: **K - E - D - A** (stored in liver and adipose tissue, hypervitaminosis possible).
> Water-soluble: **B-complex and C** (excreted in urine, regular intake required).

## 2. Infectious Diseases & Causative Pathogens Master Matrix

### A. Viral Diseases

| Disease | Causative Virus | Mode of Transmission / Key Diagnostic Feature |
|---|---|---|
| **COVID-19** | SARS-CoV-2 (enveloped positive-sense RNA) | Respiratory droplets; RT-PCR test |
| **Rabies (Hydrophobia)** | Lyssavirus / Rhabdovirus (bullet-shaped RNA) | Bite of rabid dogs, bats; affects central nervous system |
| **Poliomyelitis** | Poliovirus (Enterovirus, single-stranded RNA) | Fecal-oral route; destroys anterior horn motor neurons of spinal cord; Oral Polio Vaccine (Sabin), Inactivated Polio Vaccine (Salk) |
| **AIDS** | Human Immunodeficiency Virus (HIV - Retrovirus with Reverse Transcriptase) | Blood, sexual, maternal; attacks **CD4+ Helper T-cells**; ELISA test, Western Blot confirmatory |
| **Dengue (Break-bone fever)** | Flavivirus (DEN-1 to DEN-4, ssRNA) | Transmitted by **Aedes aegypti** mosquito (day-biting); severe thrombocytopenia (platelet drop) |
| **Chikungunya** | Chikungunya virus (Alphavirus) | Transmitted by **Aedes aegypti** and *Aedes albopictus*; severe joint pain |
| **Yellow Fever** | Flavivirus | *Aedes aegypti* mosquito vector |
| **Zika Virus** | Flavivirus | *Aedes* mosquito; causes **Microcephaly** in newborns |
| **Hepatitis B** | Hepatitis B Virus (HBV - **DNA virus**, whereas Hep A, C, D, E are RNA) | Blood, body fluids; attacks liver; vaccine available |
| **Smallpox** | Variola virus (DNA virus) | Completely **eradicated from world in 1980** (WHO) |
| **Measles, Mumps, Rubella** | Morbillivirus, Paramyxovirus, Rubivirus | MMR vaccine |

### B. Bacterial Diseases

| Disease | Causative Bacterium | Key Clinical Feature / Diagnostic Test |
|---|---|---|
| **Tuberculosis (TB)** | *Mycobacterium tuberculosis* (acid-fast bacillus) | Affects lungs; **Mantoux test**; **BCG vaccine** (Bacillus Calmette-Guérin) |
| **Cholera** | *Vibrio cholerae* (comma-shaped) | "*Rice water stools*", massive dehydration; fecal-oral route |
| **Typhoid (Enteric fever)** | *Salmonella typhi* | Step-ladder fever pattern, intestinal perforation; **Widal test** |
| **Tetanus (Lockjaw)** | *Clostridium tetani* (anaerobic, drumstick-shaped) | Neurotoxin tetanospasmin causes severe muscle spasms; ATS / Tetanus toxoid |
| **Diphtheria** | *Corynebacterium diphtheriae* | Grey pseudomembrane over throat / larynx; **Schick test** |

| Disease | Causative Bacterium | Key Clinical Feature / Diagnostic Test |
|---|---|---|
| **Pertussis (Whooping Cough)** | *Bordetella pertussis* | Paroxysmal coughing ending in whoop; DPT triple vaccine |
| **Plague (Black Death)** | *Yersinia pestis* | Transmitted by **Xenopsylla cheopis** (oriental rat flea); Bubonic, Pneumonic forms |
| **Leprosy (Hansen's Disease)** | *Mycobacterium leprae* | Hypopigmented skin patches with loss of sensation; Lepromin test |
| **Anthrax** | *Bacillus anthracis* (spore-forming) | Zoonotic disease of cattle; potential bioweapon |
| **Syphilis & Gonorrhea** | *Treponema pallidum* & *Neisseria gonorrhoeae* | Sexually transmitted bacterial infections |

### C. Protozoan & Other Pathogen Diseases

- **Malaria:** Caused by protozoan **Plasmodium** (*P. vivax, P. falciparum* [deadliest malignant tertian], *P. malariae, P. ovale*). Vector: **Female Anopheles mosquito**. Infective stage to humans: **Sporozoite**. Parasite multiplies in liver cells and destroys red blood cells (**RBCs**), releasing **hemozoin toxin** causing periodic shivering chills and high fever. Discovered by **Sir Ronald Ross** (in Secunderabad, India - Nobel Prize 1902). Treatment: Quinine (extracted from **Cinchona bark**), Artemisinin.
- **Kala-Azar (Visceral Leishmaniasis / Dum-Dum Fever):** Caused by *Leishmania donovani*. Vector: **Phlebotomus sandfly**. Enlarges spleen and liver (hepatosplenomegaly).
- **Sleeping Sickness (African Trypanosomiasis):** Caused by *Trypanosoma brucei*. Vector: **Tsetse fly** (*Glossina*).
- **Amoebic Dysentery:** Caused by *Entamoeba histolytica*.
- **Ringworm (Tinea):** Fungal infection (*Trichophyton, Microsporum, Epidermophyton*).
- **Filariasis / Elephantiasis:** Caused by parasitic nematode worm *Wuchereria bancrofti*. Transmitted by **Culex mosquito**. Blocks lymphatic vessels causing swelling of limbs/scrotum.

## 3. Human Physiology & Endocrine Hormones

| Gland | Hormones Produced | Primary Function & Prelims Exam Triggers |
|---|---|---|
| **Pituitary (Master Gland)** | Growth Hormone (GH), TSH, ACTH, FSH, LH, Prolactin, **Oxytocin**, **Vasopressin (ADH)** | Regulated by Hypothalamus. Hypersecretion of GH causes **Gigantism / Acromegaly**; hyposecretion causes **Dwarfism**. **ADH (Antidiuretic Hormone)** controls water reabsorption in kidneys; deficiency causes **Diabetes Insipidus** (frequent urination without sugar). **Oxytocin** is the milk-ejection & birth hormone. |
| **Thyroid** | **Thyroxine ($T_4$)**, Triiodothyronine ($T_3$), Calcitonin | Largest endocrine gland; requires **Iodine**. Deficiency causes **Simple Goitre** (enlarged neck), **Cretinism** in infants, **Myxedema** in adults. Hyperthyroidism causes **Grave's Disease** (exophthalmic goitre). |
| **Parathyroid** | Parathyroid Hormone (PTH / Collip's hormone) | **Increases blood calcium levels** (hypercalcemic) by dissolving bone; acts antagonistically to thyroid calcitonin. |
| **Pancreas (Islets of Langerhans)** | **Insulin** (from $\beta$-cells), **Glucagon** (from $\alpha$-cells), Somatostatin (from $\delta$-cells) | **Insulin lowers blood glucose** by converting glucose to glycogen (glycogenesis). Deficiency causes **Diabetes Mellitus** (hyperglycemia, glycosuria). Discovered by Banting & Best (1921). Glucagon raises blood glucose. |
| **Adrenal Gland** | Adrenaline (Epinephrine) & Noradrenaline; Cortisol, Aldosterone | Adrenaline is the **"Emergency Hormone" / "Fight, Flight or Fright" (3F/4S hormone)**; increases heart rate, breathing, blood pressure. |
| **Pineal Gland** | **Melatonin** | Regulates circadian rhythm (24-hour sleep-wake cycle). |

**Blood Groups & ABO System:** Discovered by **Karl Landsteiner** (1900).
- **Universal Donor: $O^-$ (O Negative)** — has neither A nor B antigens and no Rh factor on RBCs.
- **Universal Recipient: $AB^+$ (AB Positive)** — has both A and B antigens and Rh factor; has no anti-A or anti-B antibodies in plasma.
- **Normal Blood Pressure: 120/80 mm Hg** (Systolic/Diastolic), measured by **Sphygmomanometer**. Blood pH: **7.35–7.45 (slightly basic)**.

## 4. Optics & Physics Principles (High-Frequency Exam Traps)

| Phenomenon | Underlying Optical Principle | Exam Application / Examples |
|---|---|---|
| **Mirage in deserts / Looming in cold** | **Total Internal Reflection (TIR)** | Optical illusion; light travels from denser to rarer medium at angle $>$ critical angle. |
| **Sparkling of Diamond** | **Total Internal Reflection (TIR)** | Very small critical angle of diamond ($\approx 24.4^{\circ}$) due to high refractive index ($\approx 2.42$). |
| **Endoscope / Optical Fibre Cables** | **Total Internal Reflection (TIR)** | Zero signal loss communication; invented by Narinder Singh Kapany. |
| **Blue colour of sky / Red colour of sunset** | **Rayleigh Scattering of Light** | Scattering $\propto \frac{1}{\lambda^4}$. Blue light (shorter wavelength) scatters most; red light (longest wavelength) scatters least and reaches eyes at horizon. |
| **Twinkling of Stars** | **Atmospheric Refraction** | Continuously fluctuating refractive index of atmospheric air layers. |
| **Coin appears raised / Stick bent in water** | **Refraction of Light** | Light bends away from normal when entering air from water. |
| **Rainbow Formation** | **Dispersion $+$ Refraction $+$ Total Internal Reflection** | Water droplets act as tiny prisms. Primary rainbow: 2 refractions $+$ 1 internal reflection; Secondary rainbow: 2 refractions $+$ 2 internal reflections. |
| **Soap bubble shining with brilliant colours** | **Interference of Light** | Destructive and constructive interference of light reflected from upper and lower thin film surfaces. |

### Common Eye Defects & Corrective Lenses

1. **Myopia (Short-sightedness):** Nearby objects seen clearly, distant objects blurred. Image formed **in front of retina**. Corrected by **Concave Lens** (diverging lens).
2. **Hypermetropia (Long-sightedness):** Distant objects seen clearly, nearby objects blurred. Image formed **behind retina**. Corrected by **Convex Lens** (converging lens).
3. **Presbyopia (Old age vision):** Loss of accommodation power of ciliary muscles. Corrected by **Bifocal Lens** (upper part concave for distance, lower part convex for reading).
4. **Astigmatism:** Inability to focus simultaneously on horizontal and vertical lines due to irregular cornea curvature. Corrected by **Cylindrical Lens**.

## 5. Commercial Names of Common Chemical Compounds

| Chemical / Common Commercial Name | Chemical Formula | Scientific / IUPAC Name | High-Yield Application |
|---|---|---|---|
| **Baking Soda** | $\text{NaHCO}_3$ | Sodium hydrogen carbonate | Baking, antacid, fire extinguisher |
| **Washing Soda** | $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ | Sodium carbonate decahydrate | Glass, soap, water softening |

| Chemical / Common Commercial Name | Chemical Formula | Scientific / IUPAC Name | High-Yield Application |
|---|---|---|---|
| **Caustic Soda** | $\text{NaOH}$ | Sodium hydroxide | Soap manufacturing, petroleum refining |
| **Caustic Potash** | $\text{KOH}$ | Potassium hydroxide | Soft soaps, alkaline batteries |
| **Bleaching Powder** | $\text{CaOCl}_2$ | Calcium hypochlorite | Disinfectant for water, bleaching cotton |
| **Quicklime** | $\text{CaO}$ | Calcium oxide | Cement manufacture, white washing |
| **Slaked Lime** | $\text{Ca(OH)}_2$ | Calcium hydroxide | Mortar, neutralizing soil acidity |
| **Plaster of Paris (POP)** | $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ | Calcium sulphate hemihydrate | Bone fractures, statues, false ceilings |
| **Gypsum** | $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ | Calcium sulphate dihydrate | Retarder in cement setting, soil reclamation |
| **Laughing Gas** | $\text{N}_2\text{O}$ | Nitrous oxide | Dental anaesthetic, propellant |
| **Tear Gas** | $\text{CCl}_3\text{NO}_2$ (Chloropicrin) / CS gas | 2-chlorobenzalmalononitrile | Riot control |
| **Heavy Water** | $\text{D}_2\text{O}$ | Deuterium oxide | Moderator in nuclear reactors |
| **Dry Ice** | $\text{Solid CO}_2$ | Solid carbon dioxide | Refrigerant, sublimes without melting |
| **Epsom Salt** | $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ | Magnesium sulphate heptahydrate | Laxative, bath salt |
| **Blue Vitriol** | $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ | Copper sulphate pentahydrate | Fungicide (Bordeaux mixture), electroplating |
| **Green Vitriol** | $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ | Ferrous sulphate heptahydrate | Ink manufacturing |
| **White Vitriol** | $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ | Zinc sulphate heptahydrate | Eye drops, pigments |

## 6. Major Alloys & Their Compositions

| Alloy | Major Constituents (%) | Primary Properties & Applications |
|---|---|---|
| **Brass (Pital)** | **Copper (60–70%) $+$ Zinc (30–40%)** | Utensils, musical instruments, cartridge cases (*No Tin*) |
| **Bronze (Kansa)** | **Copper (88%) $+$ Tin (12%)** | Statues, medals, coins, bells (*No Zinc*) |
| **German Silver** | **Copper (50%) $+$ Zinc (30%) $+$ Nickel (20%)** | **Contains ZERO Silver!** Utensils, artificial jewellery |
| **Stainless Steel** | **Iron $+$ Chromium (18%) $+$ Nickel (8%) $+$ Carbon (0.1–1%)** | Rust-resistant; Chromium gives corrosion resistance |
| **Solder (Tanka)** | **Lead (50%) $+$ Tin (50%)** | Low melting point; joining electrical wires |

| Alloy | Major Constituents (%) | Primary Properties & Applications |
|---|---|---|
| **Duralumin** | **Aluminium (95%) $+$ Copper (4%) $+$ Magnesium (0.5%) $+$ Manganese (0.5%)** | Light and strong; aircraft bodies, pressure cookers |
| **Gun Metal** | **Copper (88%) $+$ Tin (10%) $+$ Zinc (2%)** | Heavy machinery gears, gun barrels |
| **Bell Metal** | **Copper (78%) $+$ Tin (22%)** | High resonance for casting bells |
| **Amalgam** | **Mercury $+$ Any other metal** (except Iron, Platinum, Tungsten) | Dental fillings (Silver-Mercury amalgam); **Iron does not form amalgam** |
| **Nichrome** | **Nickel (60%) $+$ Chromium (40%)** | High resistance & high melting point; electric heater coils |

## 7. Indian Space Programme (ISRO) & Defence Tech

- **ISRO Founded: 15 August 1969**; Headquarters: **Bengaluru**; Space Commission & Department of Space (DoS) formed in 1972.
- **Launch Port:** Satish Dhawan Space Centre (SDSC - SHAR), **Sriharikota (Andhra Pradesh)**, on Pulicat Lake barrier island. Second spaceport under construction at **Kulasekarapattinam (Tamil Nadu)** for SSLV launches.
- **Key Historic Missions:**
- **Aryabhata:** India's first satellite, launched on **19 April 1975** by Soviet Kosmos-3M rocket.
- **Chandrayaan-1:** Launched on **22 October 2008** via PSLV-C11; Moon Impact Probe detected water molecules ($H_2O/OH$) on lunar surface.
- **Mangalyaan (Mars Orbiter Mission - MOM):** Launched on **5 November 2013** via PSLV-C25; inserted into Martian orbit on **24 September 2014** on maiden attempt.
- **Chandrayaan-3:** Launched on **14 July 2023** via LVM3-M4; successfully achieved historic **soft landing near lunar South Pole** on **23 August 2023** (celebrated as **National Space Day**). Lander: *Vikram*, Rover: *Pragyan*. Landing site named **Shiv Shakti Point**.
- **Aditya-L1:** India's first solar observatory launched on **2 September 2023** via PSLV-C57; inserted into halo orbit around **Lagrange Point 1 (L1)** on 6 January 2024.
- **DRDO Missiles (Integrated Guided Missile Development Programme - IGMDP, 1983 - Dr. APJ Abdul Kalam):**
- **P - Prithvi:** Surface-to-Surface short-range ballistic missile (Liquid fuel).
- **A - Agni:** Surface-to-Surface medium-to-intercontinental ballistic missile (Agni I to V, Solid fuel, MIRV tech).
- **T - Trishul:** Surface-to-Air short-range missile.
- **N - Nag:** Anti-Tank guided missile ("*Fire and Forget*"); helicopter-launched version is **Helina / Dhruvastra**.
- **A - Akash:** Surface-to-Air medium-range missile with Rajendra multi-target phased array radar.
- **Nuclear Power Plants in India:**
- **Narora Atomic Power Station (NAPS): Bulandshahr, Uttar Pradesh** (PHWR, only nuclear plant in UP)
- **Tarapur:** Maharashtra (Oldest nuclear plant in India, 1969)
- **Rawatbhata:** Rajasthan (near Kota, Chambal river)
- **Kakrapar:** Gujarat (Tapi river)
- **Kaiga:** Karnataka (Kali river)
- **Kalpakkam (MAPS):** Tamil Nadu (Indira Gandhi Centre for Atomic Research)
- **Kudankulam:** Tamil Nadu (Largest nuclear power capacity in India, VVER Russian reactors)
- **Gorakhpur Nuclear Plant:** Under construction in **Fatehabad, Haryana** (Not in Gorakhpur, UP!)